

Name: Stepan, Haralambos, Chloe, Sam

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

- The system must use materials and designs that prevent mold growth to maintain water quality and plant health.

To effectively meet the needs of the stakeholders, the new water system must address key functional and performance criteria. These requirements ensure that the solution is both practical and sustainable for the greenhouse environment. Research has shown that mold-resistant materials, proper drainage, and consistent water flow are essential for a reliable irrigation system.

Design Requirement #2 (HIGH Priority)

- The system should provide consistent and uniform water delivery to all plants, ensuring optimal hydration and preventing water stagnation.

Stakeholders, including environmental science teachers and maintenance staff, agreed that mold resistance and even water distribution are the most critical factors. They emphasized that the system must be easy to maintain and adaptable to different plant species. Based on their input, the list remains largely unchanged, but additional consideration will be given to integrating an adjustable flow mechanism for different plant hydration needs.

Design Requirement #3 (HIGH Priority)

- Proper drainage mechanisms must be in place to prevent standing water and reduce moisture retention that leads to mold

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Durability and adaptability are moderately important because they contribute to the long-term effectiveness of the solution but do not directly address the primary concern. Cost-effectiveness is a lower priority because the primary goal is to implement a functional and reliable system rather than focusing solely on affordability.

Design Requirement #4 (HIGH Priority)

- The materials used should be resistant to degradation from humidity, sunlight, and frequent water exposure.

The highest priority requirements—mold resistance, even water distribution, and efficient drainage—are essential for solving the core problem. Without these features, mold buildup will continue to be an issue, negating any improvements made to the system.

Design Requirement #5 (HIGH Priority)

- The system should allow for adjustable water flow to accommodate different plant species and soil conditions.

By prioritizing mold resistance, even water distribution, and proper drainage, the new water system will effectively address all the problems of the previous water system, and will also be more reliable in the future.

Design Requirement #6 (LOW Priority)

- While budget constraints are important, the priority is to ensure a reliable and mold-free system, even if it requires a higher initial investment.

The main stakeholders that are related to this issue are the science department, as upkeep of a more expensive system can often require more care and dedication, which could change the way the science department, as well as secondary stakeholders, such as students, interact with the water system.

Design Requirement #7 (LOW Priority)

- Ensure that the design can provide a volume of water that is more than enough for the plants.



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There should be the ability to provide more water than is needed for the amount that plants need, so that in the event that some of the water passing through the system is blocked, enough water is still provided for the plants. This concerns the science department, as they would have to have a way to provide water to the plants.

Rubric

Element C: Presentation and Justification of Solution Design Requirements						
C1. Determination of design requirements. _____ design requirements are related to prior research.	score = 5 80%-100% of	score = 4 60%-79% of	score = 3 40%-59% of	score = 2 20%-39% of	score = 1 1%-19% of	score = 0 No
C2. Requirements list. Design requirements _____.	score = 5 are listed, prioritized, clear, and detailed	score = 4 are listed, prioritized, generally clear, and detailed	score = 3 are listed, prioritized, generally clear, and somewhat detailed	score = 2 are listed and prioritized but some or all may be incomplete and/or lack specificity	score = 1 are listed, formatted, and prioritization is attempted, but these may be partial or overly general	score = 0 are not present or are too vague
C3. Quantity of high-priority design requirements. _____ of the design requirements listed are identified as high priority.	score = 5 Five	score = 4 Four	score = 3 Three	score = 2 Two	score = 1 One	score = 0 None
C4. Objective & measurable. Design requirements _____.	score = 5 are consistently objective and	score = 4 are nearly always objective and	score = 3 are generally objective and measurable	score = 2 may be only sometimes objective and/or	score = 1 are insufficiently measurable	score = 0 are not measurable

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	measurable	measurable		measurable		
	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
C5. Leading to solution. Design requirements _____.	would be highly likely to lead to a tangible and viable problem solution	would be likely to lead to a tangible and viable problem solution	have a potential to lead to a tangible and viable problem solution	have some potential to lead to a tangible and viable problem solution	have little potential to lead to a tangible and viable problem solution	cannot lead to a tangible and viable problem solution
C6. Primary stakeholder validation. There is evidence that requirements represent the needs of, and have been validated by, _____ primary stakeholder groups.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	80%-100% of	60%-79% of	40%-59% of	20%-39% of	1%-19% of	no primary stakeholder groups.